

Module 2 Lesson 3
Linear Equations in One Variable and Proportions (6.2)

Objectives

In this lesson, we will learn how to

- solve linear equations;
- solve linear equations containing fractions;
- solve proportions;
- solve problems using proportions;
- identify equations with no solution or infinitely many solutions.

1. Linear Equations

A linear equation in one variable x is an equation that can be written in the form

$$ax + b = 0,$$

where a and b are real numbers, and

$$a \neq 0.$$

Linear Equation in One Variable

A linear equation in one variable can be written as

$$ax + b = 0,$$

where $a \neq 0$.

Solving an equation in x means finding all values of x that make the equation true.

These values are called solutions or roots.

Equivalent Equations

Equivalent equations have the same solution set.
For example,

$$4x + 12 = 0$$

and

$$x = -3$$

are equivalent equations.

2. Properties of Equality

Properties of equality allow us to create equivalent equations.

Addition Property of Equality

The same real number or algebraic expression may be added to both sides of an equation without changing the solution set.
If

$$a = b,$$

then

$$a + c = b + c.$$

Multiplication Property of Equality

The same nonzero real number may multiply both sides of an equation without changing the solution set.

If

$$a = b,$$

then

$$ac = bc,$$

as long as

$$c \neq 0.$$

3. Steps for Solving a Linear Equation

Solving a Linear Equation

1. Simplify the algebraic expression on each side by removing grouping symbols and combining like terms.
2. Collect all variable terms on one side and all constants on the other side.
3. Isolate the variable and solve.
4. Check the proposed solution in the original equation.

Example 1: Solve and check:

$$2(x - 4) - 5x = -5.$$

Solution

Start with the given equation:

$$2(x - 4) - 5x = -5.$$

Use the distributive property:

$$2x - 8 - 5x = -5.$$

Combine like terms:

$$-3x - 8 = -5.$$

Add 8 to both sides:

$$-3x = 3.$$

Divide both sides by -3 :

$$x = -1.$$

Check:

$$2(-1 - 4) - 5(-1) = -5.$$

$$2(-5) + 5 = -5.$$

$$-10 + 5 = -5.$$

$$-5 = -5.$$

The solution set is

$$\{-1\}.$$

4. Linear Equations Containing Fractions

Some linear equations contain fractions.

A common strategy is to multiply both sides of the equation by the least common denominator.

Clearing Fractions

To solve an equation containing fractions:

1. Find the least common denominator, or LCD.
2. Multiply every term on both sides by the LCD.
3. Simplify and solve the resulting linear equation.
4. Check the answer in the original equation.

Example 2: Solve:

$$\frac{x}{3} + \frac{x}{2} = 10.$$

Solution

The LCD of 3 and 2 is 6.
Multiply both sides by 6:

$$6\left(\frac{x}{3} + \frac{x}{2}\right) = 6(10).$$

Distribute:

$$2x + 3x = 60.$$

Combine like terms:

$$5x = 60.$$

Divide by 5:

$$x = 12.$$

5. Proportions

A **proportion** is an equation stating that two ratios are equal.

For example,

$$\frac{a}{b} = \frac{c}{d}$$

is a proportion, where

$$b \neq 0$$

and

$$d \neq 0.$$

Cross Products

If

$$\frac{a}{b} = \frac{c}{d},$$

then

$$ad = bc.$$

This is called the cross-products property.

6. Solving Proportions

To solve a proportion, use cross products and then solve the resulting equation.

Example 3: Solve the proportion:

$$\frac{x}{5} = \frac{12}{20}.$$

Solution

Use cross products:

$$20x = 5(12).$$

$$20x = 60.$$

Divide by 20:

$$x = 3.$$

Example 4: Solve the proportion:

$$\frac{63}{x} = \frac{7}{5}.$$

Solution

Use cross products:

$$63(5) = 7x.$$

$$315 = 7x.$$

Divide by 7:

$$x = 45.$$

Example 5: Solve the proportion:

$$\frac{20}{x - 10} = \frac{30}{x}.$$

Solution

Use cross products:

$$20x = 30(x - 10).$$

Distribute:

$$20x = 30x - 300.$$

Subtract $30x$ from both sides:

$$-10x = -300.$$

Divide by -10 :

$$x = 30.$$

7. Application: Using Proportions

Proportions are useful in many real-world situations, such as maps, scale drawings, recipes, and similar figures.

Example 6: A map uses a scale of

$$1 \text{ inch} = 25 \text{ miles.}$$

If two cities are 3.5 inches apart on the map, how far apart are they in real life?

Solution

Set up a proportion:

$$\frac{1}{25} = \frac{3.5}{x}.$$

Use cross products:

$$x = 25(3.5).$$

$$x = 87.5.$$

The cities are

$$87.5$$

miles apart.

Example 7: The property tax on a house with an assessed value of \$480,000 is \$5760. Determine the property tax on a house with an assessed value of \$600,000, assuming the same tax rate.

Solution

Let

x = the tax on the \$600,000 house.

Set up a proportion comparing tax to assessed value:

$$\frac{5760}{480,000} = \frac{x}{600,000}.$$

Use cross products:

$$480,000x = 5760(600,000).$$

$$480,000x = 3,456,000,000.$$

Divide both sides by 480,000:

$$x = 7200.$$

The property tax on the \$600,000 house is \$7200.

8. Application: Responding to Negative Life Events

Linear equations can be used to model real-world situations.

For example, a model may describe the relationship between the intensity of a negative life event and the average level of depression.

Example 8: Suppose the average level of depression for a high-humor group is modeled by

$$D = 0.25x + 1.5,$$

where x is the intensity of a negative life event.

Find the intensity x when the average level of depression is

$$D = \frac{7}{2}.$$

Solution

Substitute $D = \frac{7}{2}$:

$$\frac{7}{2} = 0.25x + 1.5.$$

Write decimals as fractions:

$$\frac{7}{2} = \frac{1}{4}x + \frac{3}{2}.$$

Subtract $\frac{3}{2}$ from both sides:

$$\frac{7}{2} - \frac{3}{2} = \frac{1}{4}x.$$

$$\frac{4}{2} = \frac{1}{4}x.$$

$$2 = \frac{1}{4}x.$$

Multiply both sides by 4:

$$x = 8.$$

9. Equations with No Solution

Some equations lead to a false statement.

When this happens, the equation has no solution.

Example 9: Solve:

$$2x + 6 = 2(x + 4).$$

Solution

Use the distributive property:

$$2x + 6 = 2x + 8.$$

Subtract $2x$ from both sides:

$$6 = 8.$$

This statement is false.

Therefore, the equation has no solution.

The solution set is

$$\emptyset.$$

10. Equations with Infinitely Many Solutions

Some equations lead to a true statement, such as

$$6 = 6.$$

When this happens, every real number is a solution.

Example 10: Solve:

$$4x + 6 = 6(x + 1) - 2x.$$

Solution

Use the distributive property:

$$4x + 6 = 6x + 6 - 2x.$$

Combine like terms:

$$4x + 6 = 4x + 6.$$

Subtract $4x$ from both sides:

$$6 = 6.$$

This statement is true for every value of x .
Therefore, the solution set is

$$\{x \mid x \text{ is a real number}\}.$$

11. Summary of Types of Linear Equations

Possible Results

When solving a linear equation, one of three things can happen:

Result	Meaning	Solution Set
$x = a$	One solution	$\{a\}$
False statement	No solution	$\emptyset$
True statement	Infinitely many solutions	$\mathbb{R}$

Summary

Key Ideas

- A linear equation in one variable can be written as $ax + b = 0$, where $a \neq 0$.
- Solving an equation means finding values that make the equation true.
- Equivalent equations have the same solution set.
- The addition and multiplication properties of equality help us solve equations.
- To solve equations containing fractions, multiply by the LCD.
- A proportion states that two ratios are equal.
- Cross products can be used to solve proportions.
- Some equations have no solution.
- Some equations are true for every real number and have infinitely many solutions.